

# Build the harness. Then test the models.

- I approached the brief as a harness problem: connect model APIs to a controlled Slack environment.
- I needed to understand how agents observe a page, act, and know when a task is complete.

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## 01 · RESEARCH



### How can agents act?

Screenshots, page controls, and APIs

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## 02 · BUILD



### How can I test them?

Reset, observe, act, check, record

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## 03 · COMPARE



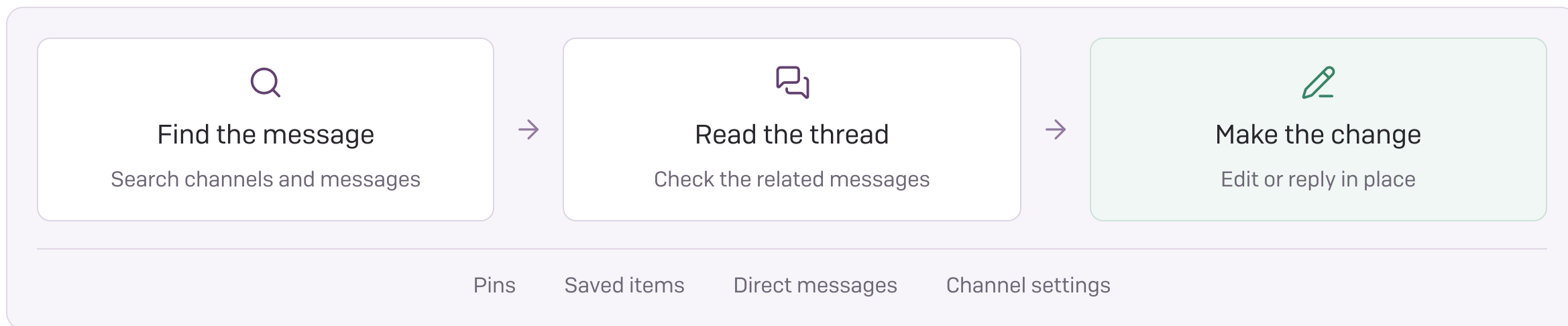
### Which models work?

Task outcomes, time, cost, and traces

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- Kevin Liu · CUA implementation exercise · October 2026 · Based on the supplied assignment brief.

# Build a small set of complete workflows

- I built the app with fictional people, channels, and messages.
- I chose search, threads, and message editing as the core workflows.
- People and browser agents use the same interface.



- React UI · SQLite state · [Try the workspace without a key](#) · [Prior-work review](#).

# 18 tasks use the full workspace

## Six focused tasks

### Update a topic

Channel → edit topic → save

### Reply in a thread

Find request → open thread → reply

### Edit a message

Find own message → edit → save

### Triage an incident

Find active incident → react → pin

### Delete a draft

Find own draft → delete → confirm

### Send a handoff

Read current details → compose DM

## Coordinate work

### Coordinate a release \*

Read sources → edit → reply → topic → DM

### Close out an incident

Read resolution → replace pin → reply → topic

### Refresh saved work

Find current sources → replace saved items → DM

### Record a design decision

Read approval → react → pin → save → details

### Repair an outdated handoff

Read correction → edit DM → react → save

### Complete a QA sign-off

Read test results → reply → fix reaction → save

## Combine several changes

### Publish a release update

Read source → format update → pin → delete draft

### Brief the on-call engineer

Read rotation → DM primary → save → details

### Correct a threaded estimate

Read review → edit own reply → react → reply

### Refresh the launch board

Find plan → replace pin → save → topic

### Hand off an approved design \*

Read decision → react → DM → reply → save

### Prepare a release retrospective

Read three sources → summarize → pin → save → DM

- These diagrams summarize the intended workflows. Exact instructions and preservation checks remain in each task.
- \* Two original tasks contain wording problems. Their results remain unchanged and are discussed later.

# Check the final workspace state

- Workspace state is the stored data: messages, threads, topics, and saved items.
- The checker compares this data with the task requirements.
- A model's claim that it finished does not affect the result.

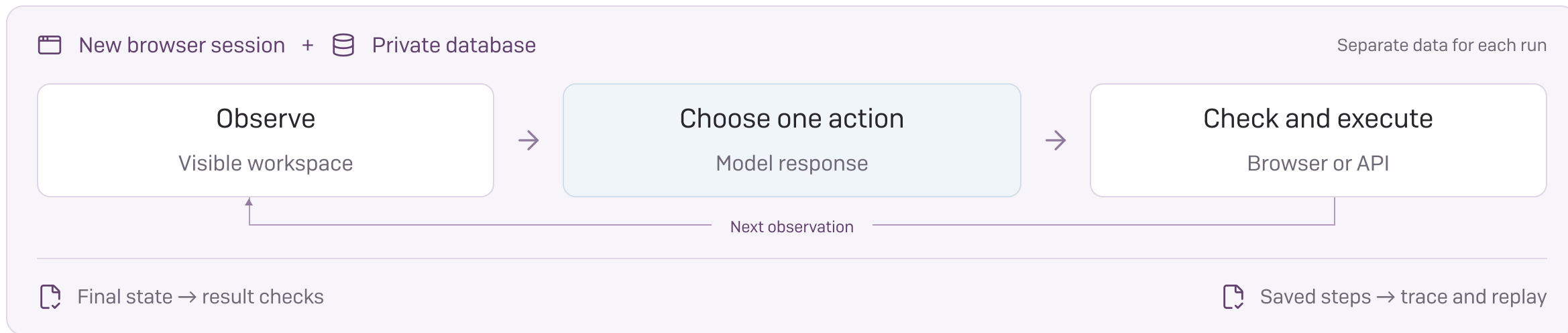
 **Example: update a channel topic**  
Set the requested topic. Make no other changes.

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|                                                                                                                                          |                                                                                                                                                   |                                |
|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
|  <b>Correct topic</b><br>Exact text in channel settings |  <b>No extra changes</b><br>Other workspace data stays the same | <b>Both pass</b><br>Reward = 1 |
|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|

- The checker grades the final state. It does not penalize a wrong action that the agent fully undoes.
- [Task definitions and result checks.](#)

# Start each run in a fresh workspace



- The harness is the code that manages each run.
  - It resets the workspace, sends observations, executes actions, and records results.
  - Each run has its own browser session and database.
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- Playwright controls the browser. Separate data does not protect the host from malicious code.

# I then used the harness to compare models

- The same run process worked with different models.
- I used it to compare 17 models on 18 tasks.
- This expanded the project into a benchmark prototype.

## Original assignment



### Working environment

Focused Slack workflows  
Known starting state



### Repeatable harness

Reset · act · check · record  
Select a model



### Benchmark prototype

18 tasks × 17 models  
One attempt per pair

The comparison also needs clear tasks and consistent test conditions.

# Each interface gives the agent different controls

| Interface                                                                                              | What the agent receives and uses   | Earlier passes / attempts |
|--------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------|
|  <b>Pixels</b>        | Screenshot → mouse and keyboard    | <b>0 / 1</b>              |
|  <b>Accessibility</b> | Named controls → element actions   | <b>2 / 6</b>              |
|  <b>Page JSON</b>     | Controls as JSON → element actions | <b>2 / 7</b>              |
|  <b>API</b>           | Workspace data → direct changes    | <b>4 / 6</b>              |

- Earlier runs used different tasks and models, so they cannot show which interface works best.
- The larger comparison uses only accessibility controls.

- 20 attempted: 8 passed, 11 incomplete, 1 blocked. 10 planned runs did not start.
- 126 requests · \$0.149874 estimated allowance, including missing-usage reservations · [Earlier interface evidence](#).

# Test the software and the agents separately

## APP BEHAVIOR



**174 backend tests**  
**91 browser tests**

Does the software work?

## RESULT CHECKS



**84 valid states**  
**2,583 invalid states**

Does grading reject wrong results?

## AGENT BEHAVIOR



**306 recorded runs**

Can the agent complete the task?

- Scripted browser tests confirm that the app supports each workflow.
- Known valid and invalid states test the checker.
- Neither check proves that a new reader can understand the task.

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- [Software verification](#) • Model evidence remains separate from scripted tests.

# Run each task once with each model

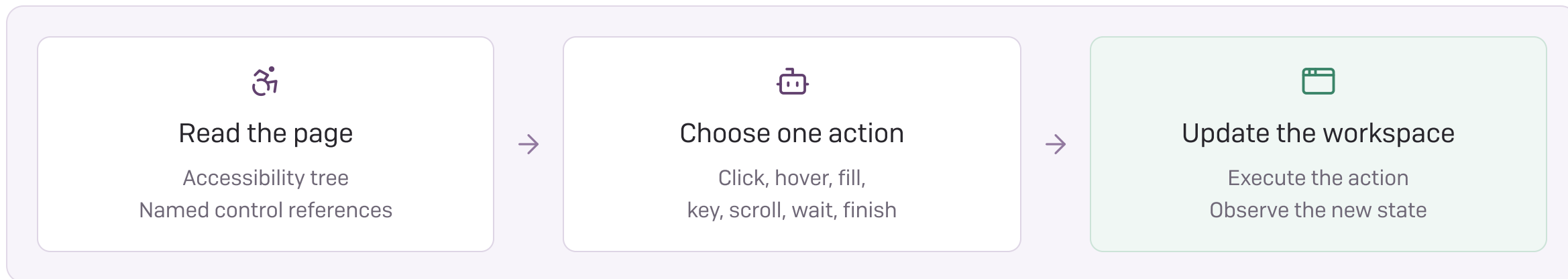
$$\begin{array}{ccccccc} \text{🏠} & & \text{☑️} & & \text{🛡️} & & \\ 17 & \times & 18 & \times & 1 & = & 306 \\ \text{model routes} & & \text{tasks} & & \text{attempt per pair} & & \text{recorded attempts} \end{array}$$

- I used six simple tasks and 12 tasks with several steps.
- All models use the same task data, action format, and result checks.
- One run per pair cannot show how often a model will succeed.

40 actions    180 s per run    \$5 estimated allowance per run

- I kept 105 earlier runs and added 201 new runs. I did not replace failed runs.
- [Node 24 setup and API key instructions](#) · [Exact routes, limits, and run history](#).

# All 306 runs used accessibility controls



- All runs used seed 1042, the last four turns, and no site guide.
- All runs allowed 40 actions, 180 seconds, and 4,096 output tokens per request.
- These results compare model routes under one interface, not which interface works best.

- 
- Checked against all 306 hash-verified episode records. Reasoning used provider defaults; routing and collection time were not controlled.
  - No screenshot-only condition, direct API condition, or llms.txt comparison is included in these 306 runs.

Results for 17 models

1 per task/model · Collection complete

306 / 306 attempted · 152 passed · 61 incomplete · 93 blocked

\$189.3110

Usage estimate

\$3.7794

Unresolved · 35 calls

\$193.0904

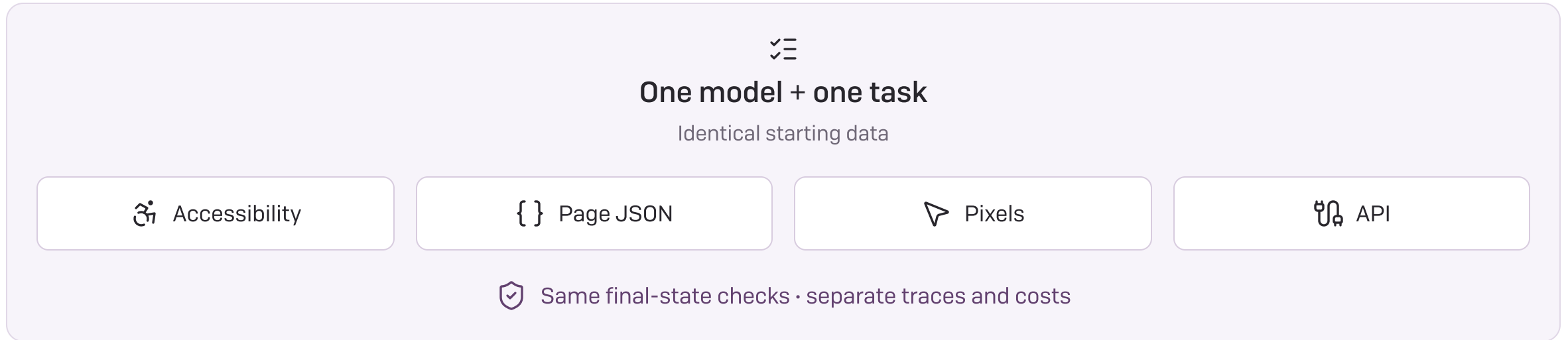
Total allowance · 306 trials

| Model ↕                                                                                               | Trials ↕ | Passed ↕ | Incomplete ↕ | Blocked ↕ | Median time ↕ | Usage estimate ↕ | Unresolved ↕ | Total allowance ↕ |
|-------------------------------------------------------------------------------------------------------|----------|----------|--------------|-----------|---------------|------------------|--------------|-------------------|
|  GPT-6 Astra         | 18 / 18  | 14 / 18  | 1            | 3         | 27.1s         | \$35.9623        | \$0.0000     | \$35.9623         |
|  GPT-6.1 Sol         | 18 / 18  | 14 / 18  | 4            | 0         | 28.5s         | \$8.4352         | \$0.0000     | \$8.4352          |
|  GPT-6 Luna          | 18 / 18  | 9 / 18   | 8            | 1         | 39.4s         | \$0.5394         | \$0.0092     | \$0.5486          |
|  GPT-OSS 120B        | 18 / 18  | 5 / 18   | 2            | 11        | 180.1s        | \$0.6682         | \$0.1358     | \$0.8040          |
|  Claude Fable 5.1    | 18 / 18  | 13 / 18  | 1            | 4         | 62.8s         | \$48.4714        | \$0.0000     | \$48.4714         |
|  Claude Opus 5.5     | 18 / 18  | 13 / 18  | 4            | 1         | 38.3s         | \$28.6987        | \$0.3741     | \$29.0728         |
|  Claude Sonnet 5.5   | 18 / 18  | 7 / 18   | 4            | 7         | 39.9s         | \$13.9323        | \$0.3304     | \$14.2627         |
|  Claude Haiku 4.5    | 18 / 18  | 6 / 18   | 3            | 9         | 28.3s         | \$3.7921         | \$0.0000     | \$3.7921          |
|  Grok 4.7            | 18 / 18  | 10 / 18  | 6            | 2         | 34.3s         | \$11.9101        | \$0.3809     | \$12.2910         |
|  Qwen 3.8 Max        | 18 / 18  | 7 / 18   | 1            | 10        | 125.1s        | \$6.9999         | \$1.0639     | \$8.0638          |
|  DeepSeek V4 Pro     | 18 / 18  | 8 / 18   | 1            | 9         | 106.8s        | \$4.6816         | \$0.2547     | \$4.9363          |
|  DeepSeek V4.1 Flash | 18 / 18  | 7 / 18   | 3            | 8         | 65.9s         | \$1.3825         | \$0.0541     | \$1.4365          |
|  GLM 5.3             | 18 / 18  | 7 / 18   | 0            | 11        | 125.3s        | \$3.6307         | \$0.3453     | \$3.9760          |
|  GLM 5.3 Flash       | 18 / 18  | 8 / 18   | 0            | 10        | 46.2s         | \$0.4444         | \$0.0000     | \$0.4444          |
|  Kimi K3             | 18 / 18  | 11 / 18  | 3            | 4         | 50.0s         | \$14.0051        | \$0.7958     | \$14.8009         |
|  MiniMax M3          | 18 / 18  | 5 / 18   | 10           | 3         | 25.9s         | \$1.4698         | \$0.0353     | \$1.5052          |
|  Nemotron 3 Ultra   | 18 / 18  | 8 / 18   | 10           | 0         | 50.1s         | \$4.2874         | \$0.0000     | \$4.2874          |

Passed Incomplete Blocked Not run

- 18 public tasks. 1 run per task and model. All use accessibility controls. Median time includes blocked runs.
- Total allowance adds the usage estimate to money reserved for missing usage. These are estimates, not invoices. Including earlier tests: \$194.1058 of the \$300 budget. [Full costs CSV](#) · [JSON](#)
- 105 earlier runs are included. No retries. One run per task cannot establish a reliable model ranking. Two tasks have unclear instructions. [Read the failure analysis.](#)
- [Test plan](#) · [Recorded runs](#) · [Download all trials \(CSV\)](#) · [Earlier 36-trial pilot: 20 passed](#)

# Change the interface, keep the task fixed



- Four models attempt six tasks through four interfaces, giving 96 planned runs.
- Each attempt starts fresh, and the interface order is randomized within each model and task.
- The study uses a separate \$25 allowance with no retries.

- 
- Sol, Sonnet, Qwen and Grok • topic, thread reply, message edit, incident closeout, saved cleanup, decision record.
  - Seed 2042 • low reasoning • recent-four history • no guide • 40 actions • 180 seconds • \$1 ceiling per cell.

# Interface comparison: 16 of 96 recorded

Passed / attempted · six planned tasks per cell

| Model                                                                                               | Accessibility                         | Page JSON                             | Pixels                                | Actor API                             | Allowance |
|-----------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|-----------|
|  GPT-6.1 Sol       | <u>1 / 1</u><br>0 blocked · 5 not run | <u>0 / 1</u><br>1 blocked · 5 not run | <u>0 / 1</u><br>1 blocked · 5 not run | <u>1 / 1</u><br>0 blocked · 5 not run | \$0.3874  |
|  Claude Sonnet 5.5 | <u>1 / 1</u><br>0 blocked · 5 not run | <u>1 / 1</u><br>0 blocked · 5 not run | <u>1 / 1</u><br>0 blocked · 5 not run | <u>1 / 1</u><br>0 blocked · 5 not run | \$0.3203  |
|  Qwen 3.8 Max      | <u>1 / 1</u><br>0 blocked · 5 not run | <u>1 / 1</u><br>0 blocked · 5 not run | <u>0 / 1</u><br>1 blocked · 5 not run | <u>1 / 1</u><br>0 blocked · 5 not run | \$0.3063  |
|  Grok 4.7          | <u>1 / 1</u><br>0 blocked · 5 not run | <u>1 / 1</u><br>0 blocked · 5 not run | <u>0 / 1</u><br>1 blocked · 5 not run | <u>1 / 1</u><br>0 blocked · 5 not run | \$0.4275  |

[All study traces and replays ↗](#)   [Costs CSV ↓](#)

- Compare interfaces within the same model and task, not against the earlier 306 runs.
  - API actions change structured data directly, so API results measure tool use.
  - One attempt per condition cannot establish a reliable winner.
- 
- Verified snapshot: 12 passed · 0 incomplete · 4 blocked · 80 not run. Usage estimate \$1.0874 + unresolved \$0.3540 = \$1.4414 of \$25. Updated 2026-10-05 06:38 UTC.
  - Pixels use low-detail screenshots. Interfaces differ in visible information and action size, not just formatting.

# The checks and the instructions must agree

- I tested the checker, but I did not ask an independent reader to solve every task.
- Two tasks had unclear instructions, so their scores do not measure model ability alone.

TASK TEXT

**“DESIGN”**

No instruction to replace this word

RESULT CHECK

≠

**“Willow”**

Checker requires this exact name

## Release sync

Unclear source location  
0 / 17 passed

## Retrospective

Every run reached a limit  
0 / 17 passed

## Handoff repair

Sol and Astra clicked Cancel  
5 / 17 passed

- Six completed runs copied DESIGN. The task did not explain that this word was a placeholder.
- Best individual models: 14 / 18 tasks. Across all models: 15 / 18 tasks passed at least once.
- Recorded actions and failure analysis. Keep all original outcomes. A wording fix requires a new task version.

# Measure local speed before adding more workers



**17.34** ms

Create session • median



**1.50** ms

Reset state • median

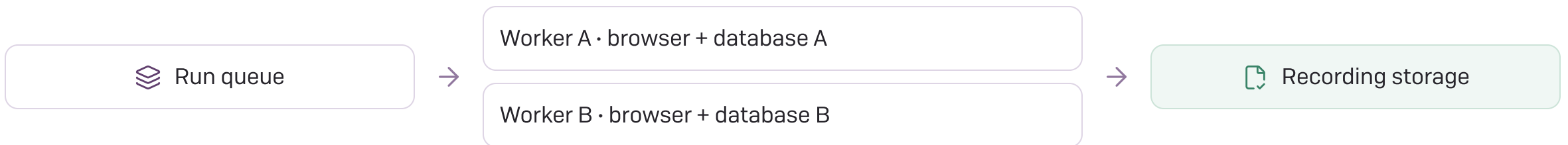


**84.15** ms

Screenshot • median

- These timings exclude model calls and do not measure how many agents can run at once.
- Next, measure memory use, completed runs per minute, and cleanup with several workers.

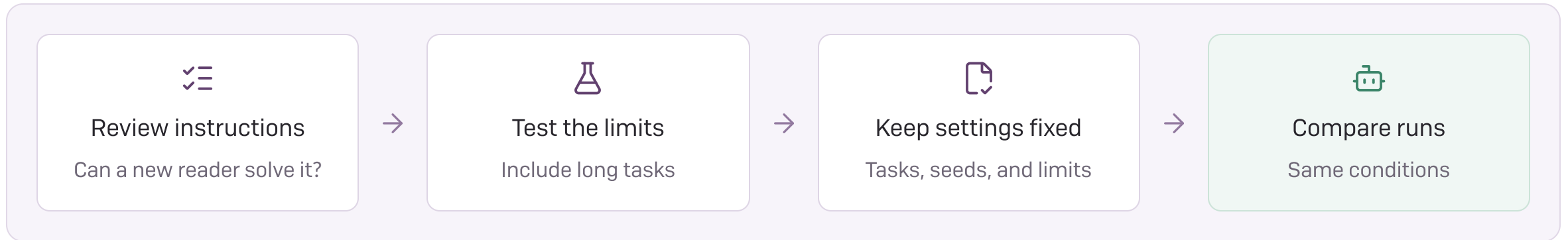
## PROPOSED DESIGN



- Apple M5 Max • Node 24 • 100 sessions • 400 reads at concurrency 16 • 10 sequential browser sessions.
- Timing samples and test setup. The multi-worker design is proposed, not measured.

# Validate each task before the next comparison

- Ask someone outside the project to solve each task from its instructions.
- Compare their interpretation with the expected result.
- Set run limits with both short and long tasks.
- Use new task variants and repeat runs to measure reliability.



Validate the task instructions as carefully as the grading code.

- [Detailed engineering review](#) · [Remaining validation work](#). These are next steps, not completed results.

# Use the app and inspect a recorded run



01 · TRY

## Use the workspace

Search, open a thread, edit



02 · COMPARE

## Choose a model and task

All 306 attempts and costs



03 · REVIEW

## Inspect the run

Actions, changes, result checks

- Which workflows should the next version test?
- Should the score include wrong actions that the agent later undoes?
- How many agents must run at once, and what should each run cost?

- 
- Relay can reset, execute actions, and return a reward. No reinforcement learning training has been run.
  - [Code and setup](#) · [Requirements and remaining work](#).